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Carbon trading and sequestration in agroforestry and cropland management: a policy perspective

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Abstract

Climate change mitigation requires innovative approaches across all economic sectors, and agriculture plays a pivotal role in carbon trading and sequestration. The carbon market plays a critical role in climate change mitigation while also providing financial incentives. The Clean Development Mechanism, a predecessor of all carbon credit schemes, has mostly focused on the compliance mechanism. Developing countries such as India, with vast lands under agrarian landholdings, need to seize the opportunities presented by the evolving voluntary carbon market (VCM). However, this potential remains largely unexplored, partly due to a lack of awareness and reluctance stemming from the procedural hurdles in securing VCM projects. The study revealed significant implementation challenges, including high upfront costs, extended payback periods of 5–10 years, limited market access for smallholder farmers, and knowledge gaps in carbon sequestration practices. This study examines the potential of carbon trading and sequestration within agroforestry and cropland management systems as market-based solutions to reduce greenhouse gas emissions and provide economic incentives to farmers. Technological innovations, including satellite monitoring, AI-based carbon accounting tools, and blockchain verification systems, have revolutionize carbon measurements, reporting and verification, and trading transparency. This study also presents a comprehensive policy roadmap that emphasizes the integration of traditional farming practices with modern carbon-efficient models. Key recommendations include strengthening policy frameworks such as the National Agroforestry Policy, establishing simplified measurement and verification systems for smallholder farmers, creating local carbon aggregator networks, and promoting indigenous land use practices. Success requires collaborative efforts involving government agencies, research institutions, and farmers, supported by streamlined registration processes, financial incentives, and comprehensive training programs, to unlock the full potential of agricultural carbon sequestration for climate change mitigation.

Keywords Climate change mitigation, Ecosystem services, Soil organic carbon, Sustainable agriculture, Voluntary carbon markets



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1 Introduction

The urgency of achieving climate neutrality necessitates a comprehensive approach across all economic sectors, demanding both a reduction in emissions and an enhancement of greenhouse gas capture in natural sinks [1]. Agricultural activities such as crop and livestock production, usually practiced on an industrial scale were responsible for approximately 11% of global greenhouse gas (GHG) emissions. Agriculture is also the largest source of anthropogenic methane, a gas over 20 times more potent than carbon dioxide [1]. Emissions from large-scale agricultural practices (crop production, livestock and fisheries) remains largely unaccounted for in compliance mechanism even though these activities are significant contributors to emission. Therefore, agriculture—vital for sustenance—faces scrutiny over its role in climate change, highlighting the need for innovative strategies to mitigate its environmental impacts [2]. This calls for improvement in land-use practices that are climate smart and leans toward traditional methods requiring fewer resource use and therefore producing lower emissions. This policy study addresses challenges and opportunities by exploring the potential of carbon trading and sequestration within agroforestry and cropland management systems.

Carbon trading in agriculture is a market-based mechanism that incentivizes farmers to adopt sustainable practices, thereby reducing GHG emissions and enhancing carbon sequestration in soils and biomass [3]. This approach provides financial incentives through carbon credits, directly rewarding environmentally beneficial actions while contributing to broader climate change mitigation goals [4]. Fundamentally, carbon trading involves the exchange of carbon credits, treating carbon—the principal GHG—as a ‘commodity’ to allows countries or entities to stay within permitted emission levels or further reduce their GHG emissions [5]. Article 17 of the Kyoto Protocol states that nations with surplus carbon units (emissions permitted but not utilized) can sell this excess capacity to countries that exceed their emission targets, fostering a flexible, economically-driven approach to emissions reduction [6].

As defined by IPCC's Sixth Assessment Report, carbon sequestration involves storing carbon in a ‘carbon pool’—a reservoir within the Earth system where carbon resides in various chemical forms for a period of time [1]. These carbon pools include the biosphere (forests and cropland), pedosphere (soils), geosphere (underground storage in suitable geologic formations), and hydrosphere (kelp beds, plankton in wetlands, and plant organic matter in peatlands and bogs) [7]. Farmers and landowners can estimate their net carbon emissions and leverage the carbon sequestered in their land use to participate in the carbon market, offering emission offsets to countries or industries seeking to reduce their carbon footprints [8]. For example, in Agriculture, Forestry, and Other Land Use (AFOLU), carbon can be traded as a removal unit (RMU) based on land use, land-use change, and forestry (LULUCF) activities [9].

Carbon trading as a climate strategy encourages industries to reduce emissions by making pollution financially costly, thereby supporting the transition to a low-carbon economy [10]. AFOLU plays a crucial role as a carbon sink, effectively reducing CO₂ levels through sequestration while improving soil fertility, and actively contributing to combating climate change [11, 12]. International agreements such as the Kyoto Protocol's Clean Development Mechanism (CDM), the UNFCCC's Reducing Emissions from Deforestation and Forest Degradation (REDD+) [13], and the Paris Agreement, 2015 [14] emphasize timely and periodical reporting of emissions and encourage carbon trading

as a tool for achieving emission targets for developed countries [15]. These agreements operate under the principle of “common but differentiated responsibilities and respective capabilities” aiming to keep the global temperature increase “well below 2 °C above pre-industrial levels” and to pursue efforts “to limit the temperature increase to 1.5 °C above pre-industrial levels” [16].

Global carbon credit standards, such as Verra’s Verified Carbon Standard (VCS) and Gold Standard (GS), provide tradable carbon credit certifications for use in international carbon markets, ensuring the integrity and environmental benefit of carbon offset projects [17]. Despite the emergence of several such registries and new opportunities in the voluntary carbon markets (VCMs), research on the actual impacts and benefits for farmers is lacking; consequently, no statistical database yet exists on these outcomes. Countries implementing carbon pricing policies have reported reduced emissions, demonstrating the effectiveness of market-based approaches in reducing GHG emissions [18].

In an ambitious plan, the USA aims to extract one gigaton of carbon through the conversion of 35 million hectares of pure pasture land in the Great Plains into agroforestry systems via the phased introduction of native trees with high leaf-shedding rates [19]. India’s policies for carbon capture—such as the National Afforestation Programme (NAP), Green India Mission (GIM), Compensatory Afforestation Fund Management and Planning Authority (CAMPA), National Agroforestry Policy, Green Credit Programme, and Digital India Land Records Modernization Programme (DILRMP)—are expected to greatly enhance farmers’ participation in the VCM [20]. However, according to the National Forest Policy of India 1988 [21] and the Forest Rights Act 2006 [22], land-record verification is required. This requirement makes it extremely challenging for smallholder farmers to undertake carbon projects, as the digitization of agroforestry lands owned by tribals and forest communities remains far from complete. Thus, AFOLU projects suffer from complex regulatory frameworks related to land ownership and use, conflicts which are exacerbated by climate change-related natural hazards [20]. Climate change induced extreme weather events remain a cause of critical macroeconomic and financial policy challenges [23]. Smallholder farmers are not motivated to participate in VCMs because it is difficult for them to align with the methodologies of private carbon market registries. Without smallholder participation, it becomes extremely challenging to achieve carbon neutrality and fulfil Nationally Determined Contributions (NDCs). Moreover, there is growing apprehension about the future of carbon market’s future due to ever-evolving regulatory standards [24].

This policy paper will delve into the mechanisms, challenges, and opportunities associated with carbon trading and sequestration in agroforestry and cropland management, providing insights for policymakers and stakeholders seeking to leverage these strategies for climate change mitigation and sustainable land use practices. The objectives of this study are: (1) to examine the potential of agroforestry and cropland management for carbon sequestration and carbon trading; (2) to identify the best carbon accounting methods of by comparing traditional approaches with advanced techniques; (3) to understand the challenges smallholder farmers face in implementing carbon credits projects; (4) to envision carbon markets that meet the needs of smallholder farmers; and (5) to design a policy roadmap for effective carbon trading in agroforestry and cropland management.

2 Methodological approach

With regard to the development of carbon accounting and inclusion of AFOLU for the GHG inventory, rapid conversion of forest to non-forest/agricultural land-use and considering the importance of such land use in global carbon dynamics, the review was devised. The PRISMA 2020¹ guideline was adopted for preparing this literature review (Fig. 1). The period covered for this study is from 1997 with the Kyoto Protocol which promulgate the carbon market in global scale to 2025 to study latest development. Relevant literature was collected from various repositories including Google Scholar, ResearchRabbit, ScienceDirect, and Scopus. Grey literature was assessed from relevant websites such as the Food and Agriculture Organization (FAO), Gold Standard, Indian Council of Agricultural Research (ICAR), International Monetary Fund (IMF), Intergovernmental Panel on Climate Change (IPCC), Ministry of Agriculture and Farmers Welfare (MoA & FW) of the Government of India, Ministry of Environment, Forest and Climate Change (MOEFCC) of the Government of India, United Nations Development Programme, (UNDP), United Nations Framework Convention on Climate Change (UNFCCC), Verra and World Bank. The keywords used during the literature search includes advanced techniques in carbon measurement, measurements, reporting and

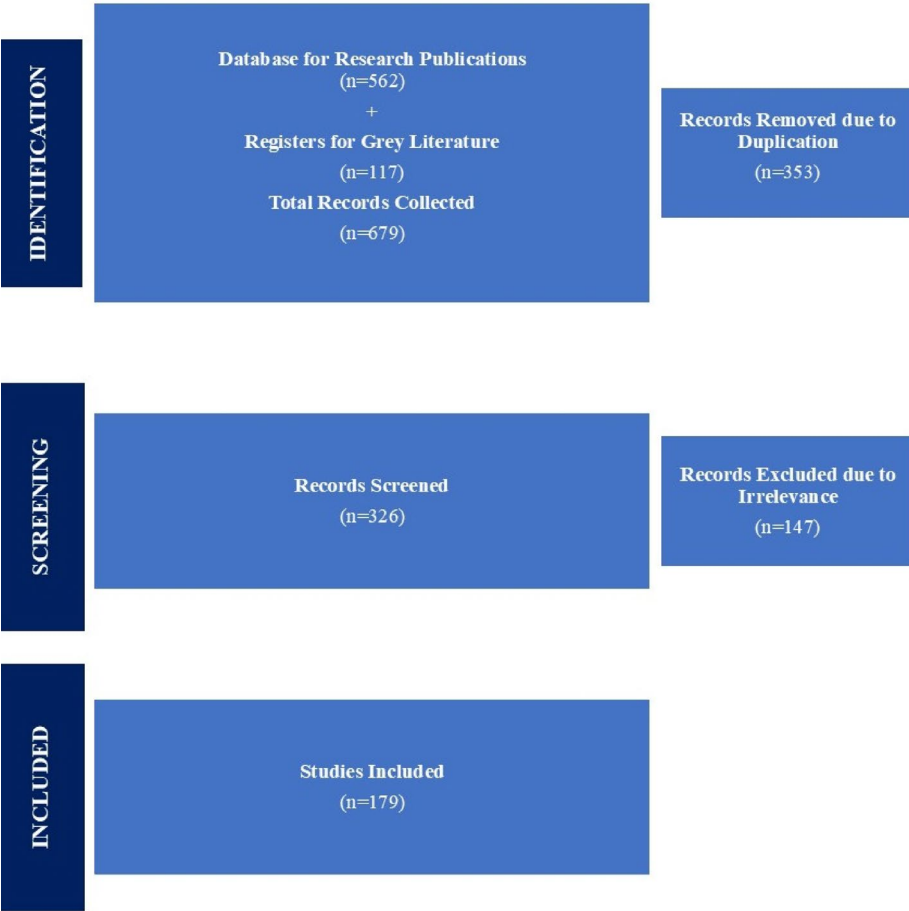


Fig. 1 PRISMA 2020 flowchart depicting Identification, Screening and Inclusion for writing manuscript

¹<https://www.prisma-statement.org/prisma-2020>.

verification (MRV) in carbon market, agroforestry, agroforestry systems, carbon credit, carbon credit projects, carbon market AND agroforestry, carbon sequestration AND compliance AND voluntary carbon market, cropland management OR agroforestry land use, ecosystem service, land use AND carbon sequestration, agricultural policy AND farmers incentives, soil organic carbon AND carbon sequestration, removal unit, sustainable agricultural practices, etc. The discussion is presented with suitable subheadings in subsequent sections.

3 Carbon sequestration in agroforestry and cropland management

Naturally, carbon sequestration captures and stores atmospheric CO₂ in plants, soil, water, and organic matter, and plays a crucial role in balancing the global carbon cycle and mitigating climate change [25]. This process involves the uptake of CO₂ by vegetation through photosynthesis, converting it into biomass and storing it in various plant parts [26]. Trees, shrubs, and grasses act as long-term carbon sinks, accumulating carbon over their lifespan [27].

Soil organic matter (SOM) is a significant carbon reservoir [28], and healthy soils store carbon in the roots, decomposed plant material, and microbial biomass [29]. Practices such as reduced tillage, organic amendments (compost, biochar, and manure), and cover cropping enhance soil carbon storage [28]. Agroforestry, which integrates trees into agricultural landscapes, further contributes to carbon sequestration in both vegetation and soil [30]. Enhancing natural carbon storage in agricultural systems is vital for mitigating climate change [28]. It helps offset emissions from AFOLU [2, 31, 32] as well as industries [2]. These practices prevent soil degradation, improve biodiversity, and promote overall ecosystem health [33]. Strategies focused on enhancing carbon inputs to soils and minimizing carbon losses are essential for maximizing the potential of agricultural lands as carbon sinks [34].

3.1 Integrating carbon sequestration into sustainable agricultural practices

Integrating ecological principles offers a comprehensive strategy for farming systems, emphasizing biodiversity, soil health, and sustainable resource management [35]. By focusing on these elements, agroecological practices can enhance carbon sequestration and improving farm resilience [36]. This holistic approach uses several key techniques to maximize carbon storage and minimize environmental impact. Soil microbial management is crucial, because it promotes beneficial microbes that aid in storing carbon in organic compounds [37]. This involves the application of biofertilizers, compost, and mycorrhizal fungi to improve carbon retention in the soil [38]. Furthermore, water-efficient irrigation practices, such as drip irrigation, contour farming, and rainwater harvesting, play a vital role in reducing soil erosion and maintaining soil moisture, thereby preventing carbon loss [39].

The benefits of agroecological carbon sequestration are multifaceted. It reduces reliance on chemical inputs, such as synthetic fertilizers that release CO₂ and N₂O, and synthetic pest control substances [40]. Enhancing biodiversity is another key advantage, as it improves soil carbon storage and overall ecosystem health [41]. Selvan et al. [32] studied peri-urban agroecological landscapes and observed a profound performance in carbon sequestration of woody components majorly constituting bamboos, totalling 362,629.68 kg CO₂e against total emission from crop at 100,086.53 kgCO₂e,

which approximately translates to 3 times. This was achieved because of the negligible use of farm inputs and lack of mechanization. Ultimately, agroecological approaches strengthen food security while mitigating climate change and conserving water, thereby minimizing farmyard runoff and prevents methane release [42].

3.2 Relevance to agroforestry and cropland management

Agroforestry and cropland management are crucial for carbon sequestration, as they offer pathways to mitigate climate change and enhance environmental sustainability. By integrating woody components into agricultural landscapes, agroforestry mimics long-lived natural forests, excelling in carbon capture and storage within roots, stems, branches, and leaves while providing multiple associated ecosystem benefits [43]. This approach combines agricultural and forestry practices to create diverse, productive, and sustainable land use, maximizing the benefits of the interactions between woody, herb/shrub crops and livestock components.

Agroforestry is practiced widely in regions such as sub-Saharan Africa (SSA), Latin America, and South and Southeast Asia, and is promoted by organizations such as FAO, IUFRO, and ICRAF as a key tool for climate change mitigation [44, 45]. Ghale et al. [46] conducted a thorough review of the carbon sequestration potential of agroforestry systems across different regions. Different publications provide different nomenclature for agroecological practices which vary greatly across regions. The study found that carbon storage potential in tree biomass across different agroforestry systems in Asia (South and Southeast Asia) ranged from just about 6.6 t C/ha/yr to 272 t C/ha/yr depending largely on the species combination with higher mixture of woody trees showing higher sequestration. In Africa the carbon sequestration potential ranged from about 9.2 t C/ha/yr up to 47.3 t C/ha/yr [46]. Therefore, different agroforestry systems offer different benefits. Agro-silviculture, which involves planting trees alongside crops, enhances soil carbon and nutrient fixation [47]. Silvi-pastoral systems combine trees with livestock grazing, to improve carbon retention and soil health [43]. Agri-silvi-pastoral systems integrate crop, trees, and livestock rearing for optimum land utilization. Alley cropping and windbreaks use woody components to prevent topsoil erosion and runoff, and effectively capture crop emissions [48]. Agroforestry in agro-industrial plantations combines cash crops such as coffee, cocoa, and tea with shade and nitrogen-fixing trees [49]. Forest farming raises crops, such as medicinal plants, spices, and edible fungi under tree canopies [50]. Woody perennials in Silvi-pastoral land use stands can sequester an average of 2.05 t C/ha/yr [51].

Cropland management strategies encompass farming practices to improve land productivity while considering sustainability and environmental health [52]. The scaled adoption of these practices can significantly reduce agricultural emissions, enhance soil fertility, and maximize carbon capture. Conservation tillage reduces soil disturbance, preserves soil structure, reduces SOC escape, improves SOM, and decreases farm emissions [52]. Cover cropping involves non-harvested crops to improve soil health, enhance soil organic matter and carbon storage, and prevent erosion [53]. Composting and biochar application enhance soil carbon storage, improve soil quality and reduce reliance on synthetic inputs [54]. Crop rotation and diversification involve sequential cultivation of crops to improve soil fertility and microbial activity, reduce pest/disease incidence, and promote carbon sequestration and nitrogen fixation [55]. Circular economy principles

are applied to croplands, particularly when using organic residues to prevent emissions, retain carbon, and reduce methane emissions [56].

Major crops, especially cereals, consume enormous amounts of energy, labour, and synthetic inputs, contributing to approximately 24% of global emissions from agriculture [57]. The global average of sequestration from croplands is 0.56 t C/ha/yr [58]. Single legume cover-crops sequester 0.41 t C/ha/yr [59], while single non-legume cover-crops sequester 0.58 t C/ha/yr [51]. Double cover cropping with one legume and one non-legume yield a high mean carbon sequestration of 1.2 t C/ha/yr [51]. Sequestration potentials in regenerative land use with woody perennials range from 1.10 to 2.05 t C/ha/yr, while in cropland, it is 0.76 t C/ha/yr [51]. The combination of cover crops and no-tillage in agroforestry and cropland results in higher SOC of 1.01 t C/ha/yr and 1.43 t C/ha/yr, respectively [51]. These practices can be used interchangeably or in combination to enhance carbon capture in AFOLU.

Apart from enhanced productivity, crop diversification, enhanced agrobiodiversity, reduced input costs, and reduced resource and energy input, the major benefit to farmers is reduced GHG emission and hence positive net emissions [60]. The adoption of sustainable practices improves productivity, increases farm income, and enhances resilience to climate change through sustained and diversified income [61]. Self-sufficiency in food and the prevention of localized farm disasters due to hydro-meteorological hazards further benefit farmers, thus avoiding huge recovery costs.

4 Carbon credits and their valuation

Carbon credits are measurable, verifiable, and tradable instruments representing one metric ton of carbon dioxide equivalent (1 t CO₂e) reduced or removed from the atmosphere [4]. These credits are typically issued to projects that actively capture, avoid, or reduce emissions, with compliance mechanisms often seen in afforestation and reforestation programs under the Joint Implementation (JI) and CDM [62]. They can also be awarded for voluntary actions undertaken by farmers and landowners, such as implementing agroforestry practices, afforestation initiatives, and improved cropland management techniques [63].

There are two primary types of carbon credit: removal credits and avoidance/reduction credits. Removal credits are generated through activities that sequester carbon, such as tree planting and enhancing soil carbon storage [64]. Avoidance/reduction credits, on the other hand, are earned by preventing emissions through conservation efforts, reducing deforestation rates, and capturing methane [65]. The valuation of carbon credits is influenced by several factors, including market demand and supply dynamics, which cause fluctuation in prices based on global demand for carbon offsets [66]. Project type and certification also play a significant role, with high-quality projects such as afforestation initiatives generally commanding higher prices in the market [67]. Additionally, projects that offer co-benefits, such as enhanced social outcomes, biodiversity conservation, and improved water conservation efforts, often receive premium pricing [68].

Carbon credit markets operate under two main frameworks: compliance carbon markets (CCMs) and voluntary carbon markets (VCMs). CCM credits, which are utilized in regulated sectors, typically range from \$10 to \$100 per ton of carbon but can exceed \$100 per ton depending on the region and specific policies [69]. VCM prices generally range between \$5 and \$50 per ton of carbon dioxide equivalent (CO₂e) [70]. CCMs are

mandatory markets where governments impose emission reduction targets, exemplified by systems such as the EU Emissions Trading System (EU ETS), the California Cap-and-Trade Program, and China's National Carbon Market [71]). Industries participating in these markets are required to purchase allowances or credits to meet their regulatory obligations [72]. Agroforestry and cropland projects can be included in some compliance systems, but are subject to stringent verification processes [63]. VCMs are open to businesses, individuals, and organizations seeking to voluntarily offset their emissions through projects such as reforestation, regenerative agriculture, and soil carbon sequestration [73]. Key registries in the VCM include Verra (Verified Carbon Standard—VCS), Gold Standard (GS), and the American Carbon Registry (ACR), providing more flexible and accessible options for farmers and landowners [67]. These options are particularly important for agroforestry and cropland management because they offer easier access for farmers to sell carbon credits, encourage the adoption of sustainable agricultural practices, and help companies achieve their corporate sustainability goals [74].

4.1 Agroforestry in global carbon markets

Agroforestry systems have gained recognition as efficient natural carbon sinks, making them attractive for the global carbon markets [26]. These systems promote tree planting and sustainable land management, thereby offering large-scale carbon sequestration [75]. The role of agroforestry extends to reducing carbon emissions from deforestation and land degradation, conserving biodiversity, and enhancing ecosystem services. The ability of agroforestry to combine climate change adaptation and mitigation provides opportunities to create synergies between these actions [76]. Major carbon trading platforms are increasingly supporting agroforestry projects. The VCS or Verra is one of the largest currencies in VCMs. The GS promotes high-impact carbon reduction projects, including agroforestry and sustainable agriculture, to provide carbon credits for trade [77]. Additionally, the CDM aims to promote carbon reduction projects in developing countries, further supporting agroforestry initiatives [78]. The CDM defines forests as areas with 0.05–1.0 hectare, >10–30% tree cover, and trees with a minimum height of 2–5 m [79]. A comparison (Table 1) for the CDM, Gold Standard, and Verra was prepared from information obtained from von Avenarius et al. [80], Gold Standard's validation and verification standard [81], principles and requirement [82], TOOL03 CDM for Measuring, Reporting, and Verification [83], Gold Standard,² UNFCCC's CDM,³ and Verra.^{4,5}

4.2 Challenges and considerations in implementing carbon credit projects

The implementation of carbon credit projects in agroforestry and cropland management presents significant hurdles that potentially diminish their overall impact and widespread acceptance. These challenges span substantial initial expenses and extended periods before returns to limited market entry, knowledge gaps, and technical complexities. Effectively addressing these issues is paramount to fully harnessing the ability to sequester carbon within agricultural settings.

² <https://www.goldstandard.org>.

³ <https://cdm.unfccc.int>.

⁴ <https://verra.org>.

⁵ <https://verra.org/programs/verified-carbon-standard/vcs/program-details/>.

One of the primary obstacles is the high cost associated with establishing agroforestry systems and implementing improved soil management practices [84]. Implementing sustainable land management practices such as agroforestry often involves high upfront maintenance costs [79]. These expenses frequently encompass tree planting, thorough land preparation, installation of efficient irrigation systems, and labor [84]. These initial outlays can be particularly prohibitive for smallholder farmers, who often face significant constraints in accessing sufficient capital for such investments [85]. The financial burden is further amplified by the necessity of specialized machinery designed for conservation tillage, placing these practices beyond the immediate reach of many farmers without external assistance [86]. Moreover, the costs associated with project registration and the consultation fees required for carbon credit certification add to the financial strain, deterring participation and reducing the economic attractiveness of these projects [87].

Another significant impediment is the long timeframe required to realize financial returns from carbon sequestration projects. In agroforestry, carbon sequestration efforts typically yield measurable carbon credits only after 5–10 years [88]. The delayed financial returns can undermine challenge farmers' long-term commitment, especially when immediate financial needs compete with environmental goals [84]. The gap between the initial investment and eventual returns necessitates the development of innovative financial instruments and incentives designed to support farmers during the interim period [89]. Without such support, the economic viability of these projects, particularly those reliant on short-term gains, remains questionable [85].

Access to carbon credit markets is frequently restricted by a confluence of factors, creating additional barriers for farmers seeking to benefit from carbon sequestration. The methodologies for verifying and accurately measuring carbon sequestration continue to evolve, leading to uncertainties in credit quantification and market value [90]. Complex registration processes, often mandated by stringent standards such as the VCS or GS, can be prohibitively time-consuming and expensive, further restricting access to smallholder farmers [87]. The absence of direct buyers forces farmers to rely on intermediaries to sell carbon credits, which reduces their profit margins and adds extra layers of complexity [91]. Fluctuations in carbon prices add another layer of uncertainty, making it difficult for farmers to predict their earnings accurately and justify the initial investments required for project implementation [92]. Limited access to affordable and reliable financing for agroforestry initiatives and carbon credit registration exacerbates these challenges, particularly for smallholder farmers who lack the resources to navigate the complexities of the carbon market [85].

Knowledge and technical gaps represent substantial barriers to adopting carbon sequestration practices in agroforestry and cropland management. Many farmers lack sufficient awareness and understanding of carbon sequestration practices, which impedes their ability to implement these techniques effectively [92]. The concept of carbon credit markets and their revenue potential are often unfamiliar to farmers, especially in regions with lower levels of education [92]. Furthermore, farmers often lack the technical expertise to accurately measure carbon sequestration or to prepare the necessary project documentation for carbon credits, which creates a significant obstacle to participation [90]. The absence of adequate training and exposure to agroforestry techniques, carbon farming techniques, carbon monitoring, and sustainable farm

Table 1 Comparison between Clean Development Mechanism (CDM), Gold Standard (GS) and Verra

Criteria	Clean Development Mechanism (CDM)	Gold Standard (GS)	Verra
Credit Units	Certified Emission Reduction (CER)	Gold Standard Verified Emission Reduction (GS-VER)	Verified Carbon Units (VCU)
Type	Compliance + Voluntary	Voluntary	Voluntary + Compliance
Monitoring	Consideration of host entity's (country) Nationally Determined Contributions (NDCs) Monitoring of SDG impacts	Preliminary Review of Key Project Information (for Project Design and Development, Eligibility, Stakeholder engagement, Scope, Applicability) for listing into Impact Registry. Impact measurements, development and performance with reference to quantification methods for climate SDGs through the SDG Impact Tool	Preliminary review of project designs submitted by stakeholders for eligibility, scope, and information of baseline assessment, GHG quantification approach Public stakeholder consultation
Reporting	Measured values should be accurate which means no overestimation nor underestimation allowed in reporting. All relevant emission and sinks must be reported	Accurate and transparent accounting and reporting must be ensured	Verification and Validation Bodies (VBs) that satisfy the <i>Methodology Submission Form and Agreement</i> selected which will also give reports for carbon measured. The VVB shall produce an assessment report in accordance with the VCS Program rules and best practices using the <i>VCS Methodology Assessment Report Template</i>
Verification	Verifiers are to be assigned by the UNFCCC Secretariat per random allotment to an activity owner or local verification experts	Gold Standard Validation and Verification Bodies, and in circumstances of applicants in conflict zones or high-risk zones on-site Validation by an Objective Observer (OO)	
Location	Across the world	Across the world	Across the world
Projects Registered	8188	3848	3000+
Carbon Credits Issued (in B)	2.5	0.4	1.8
Carbon Credits Retired (in M)	–	193	343

Table 1 (continued)

Criteria	Clean Development Mechanism (CDM)	Gold Standard (GS)	Verra
Total Shared Value Creation (in B \$)	245	59.5	–
SDGs Covered by projects registered in addition to SDG 13 (Climate Action)	SDGs 1, 3, 6, 7, 8, 9, 11, 12 are covered with majority of projects focussing on socio-economic indicators. Additional emphasis of CDM is on SDG 17 (Partnership for the Goals)	17 out of 17 SDGs are covered. Minimum of 3 SDGs always including SDG 13 are covered in all Gold Standard-certified projects. Some project may fulfil up to 16 SDGs ^a with the most frequently covered being SDG 7 (Affordable and Clean Energy)	17 out of 17 SDGs are covered with projects focussing on socio-economic indicators. Trending emphasis on SDG 7 (Affordable and Clean Energy), SDG 11 (Sustainable Cities and Communities) can be observed

B = Billion, M = Million, \$ = US Dollar, SDG = Sustainable Development Goal

^a<https://dashboard.goldstandard.org/>

management practices further compounds these challenges, limiting farmers' capacity to engage effectively in carbon credit projects [92].

Overcoming these multifaceted challenges necessitates a comprehensive strategy that encompasses supportive policies, streamlined processes, financial incentives, and extensive knowledge dissemination. Policies and incentives are vital to encourage the broad adoption of carbon sequestration practices, making them economically appealing and accessible to farmers [93]. Simplifying project registration, promoting direct buyers–farmers linkages, and ensuring stable carbon credit pricing can significantly improve market access and profitability while incentivizing greater participation [94]. The provision of subsidized loans, government grants, and financial support from relevant institutions can alleviate the financial burden on farmers, enabling them to cover the initial investment and ongoing management costs associated with carbon sequestration projects [89]. Offering well-structured training programs, accessible extension services, and consistent technical support can bridge knowledge gaps, empower farmers to participate effectively in carbon credit projects and optimize their carbon sequestration efforts [92].

5 Policies incentives and benefits for farmers and landowners

Incentives are crucial to encourage farmers and landowners to adopt carbon sequestration practices, given the high upfront implementation costs, the need for technical support to measure and verify carbon storage, and the long payback period for carbon credit revenue [95]. To overcome these barriers, a range of policy incentives can be implemented—including carbon payment programs that directly pay farmers to sequester carbon, as seen in the U.S. Climate Smart Agriculture Initiative and the EU's Common Agricultural Policy (CAP) climate payments [96]. Subsidies and grants for sustainable practices, such as no-till farming, cover cropping, and agroforestry, can reduce the costs of transitioning to carbon-friendly farming [84]. Tax credits and exemptions for carbon sequestration and capture projects provide additional financial incentives for participating in carbon offset programs.

Public–private partnerships (PPPs) can also play a significant role in supporting carbon trading in agriculture. For example, the World Bank's BioCarbon Fund supports agroforestry and cropland projects, and the Africa Carbon Market Initiative (ACMI) helps African farmers access carbon markets [84]. Carbon pricing policies—where governments set a price on carbon—encourage participation in carbon credit markets. Robust carbon prices increase the value of carbon credits, particularly for smallholder farmers in VCMs [97]. Agroforestry promotion programs, such as India's National Agroforestry Policy (2014), support the integration of trees into agricultural landscapes [97]. These policies benefit farmers and landowners by providing additional long-term revenue streams through carbon credit sales, encouraging sustainable agriculture, and supporting climate resilience by promoting regenerative farming techniques [44, 45, 98].

5.1 Future outlook for agroforestry and cropland management w.r.t carbon trading

Regulatory challenges in carbon markets will continue to evolve, but policy responses are becoming more sophisticated to meet these challenges [99]. Agroforestry and cropland management present significant opportunities for the evolving carbon market and for climate change mitigation in the coming years, as these practices can effectively sequester atmospheric carbon [100–102]. Sustainable soil carbon sequestration practices are

crucial and need to be rapidly scaled up for effective climate change mitigation [102]. The potential for carbon sequestration is particularly high in cropland soils, especially in those that have experienced historic organic carbon losses or significant yield gaps [103]. The carbon market for agroforestry and cropland management has a promising prospect for future adoption and development [104, 105] as there is growing global recognition of such land use as a sustainable solution to food security and carbon capture [106].

Carbon Credits earned domestically can be traded to meet the Nationally Determined Contributions as per Article 6 of the Paris Agreement [107]. However, the VCM requires strong governance, integrity and transparency in order to grow further. Tightening compliance caps can enhance the relevance of credits from VCMs and help keep carbon credit prices afloat [108]. This is underscored by a growing surplus of unretired credits in the market—nearly 1 billion tons issued before 2022 which are from AFOLU or renewable energy products [109]. However, this does not mean AFOLU-based credits are losing demand globally. Carbon credits from land use projects remain relatively high-priced (about \$15.5 in April 2025), whereas credits for avoided emissions (e.g. carbon capture technologies, or not cutting down trees) are around \$5.3 per unit [109].

The demand for carbon credits is projected to increase dramatically, potentially five times by 2030, as more companies commit to achieving net-zero emissions targets [103]. This surge in demand will likely drive greater participation in VCM, especially among smallholder farmers, who are becoming increasingly aware of sustainable agroecological practices and the economic benefits of carbon trading [103, 110]. Higher participation in VCMs smallholder farmers is expected, driven greater awareness of sustainable agroecological practices and increasing interest in carbon trading. However, as seen in over 80 Emission Trading Systems, a lack of price-supply adjustment mechanisms can lead to a prolonged period of low prices. For India's ICM and upcoming CCTS, integrating stability tools early on can be a vital to ensuring a stable market where carbon price does not crash. Shrivastava et al. [111] suggest a stability mechanism with three elements: "a consignment auction system for transparent, rule-based credit interventions; a vintage-based credit classification to limit the impact of older surplus credits on future prices; and an established price corridor to guide interventions when carbon prices deviate from expected levels".

Strengthening carbon markets through improved government policies is essential for enhancing the economy and ensuring the integrity and effectiveness of carbon trading schemes [112]. Policies that support carbon sequestration in agriculture can play a key role in achieving carbon neutrality goals [113]. The value of traded units is likely to increase as climate policy objectives become more ambitious and converge around net-zero emissions, inducing greater market interconnectedness as outlined in Article 6 of the Paris Agreement (this interconnectedness framework has yet to be effectively implemented) [107]. The proliferation of new digital tools for advanced MRV and remote sensing for compliance control can enhance tightening of caps and hence increase the demand for credit units in the market.

Fifteen countries signed emission reduction agreements with the World Bank to trade their high-integrity carbon credits (to be purchased by the Forest Carbon Partnership Facility), and five of them having completed their MRV received payment of \$90 M which are shared in the communities [108]. Emphasis must be placed on the integration of market mechanisms with domestic mitigation policies to reconcile

market mechanisms through robust safeguards against negative social and environmental impacts with efforts from government and non-government organizations [114, 115]. No global policy framework currently exists to standardize MRV, and one must be developed urgently to enable fair trade in the international carbon market [24]. The Global Tipping Points Report 2025 [115] highlighted that global warming will soon overshoot 1.5 °C, which demands immediate intervention with effective governance strategies that includes regeneration of nature for scalable and sustainable carbon removal and sustainable food production systems that can help eliminate 25% of GHG emissions linked to the AFOLU. Forest certification is an effective market-based mechanism for sustainable forest management that can improve forest governance and enhance adoption of carbon credit projects [116]. The simplification of access to carbon markets by policy makers and advanced techniques in quantifying the carbon sequestration values of various land use practices can enhance the adoption of carbon markets [105].

In India the VCM carbon credit prices in the early 2025 range roughly between ₹200 to ₹400 per ton of CO₂e, which is approximately \$2 to \$5 per ton. India's upcoming compliance carbon market, the Carbon Credit Trading Scheme (CCTS), set to begin in 2026, is expected to significantly influence carbon credit prices by creating a regulated and market-driven environment⁶ for emissions and credit trading across sectors. This government-supported compliance market and its related regulations are expected to stabilize and increase prices, aiming at approximately \$10 per ton as an initial benchmark in 2026.⁷

The introduction of the Gold Standard New Soil Organic Carbon Model Guidelines [117] will ease the procedure involved in the application of agroforestry and cropland managers (AFOLU sector) as VCS projects under the Gold Standard. This facilitates for project designs that is “best suited to their local geography, soil and climate and then follow a seven-step process that will cover projects’ objectives, model selection, data collection, calibration, validation, prediction and verification” [117]. When we look at the Gold Standard Report of 2024 [118], we see that the “land use and forestry” or AFOLU in various stages numbered only 75 out of 3848 total projects. However, the land use and forestry category is the fastest growing accounting for 7% of the 2024 figures, which is a significant increase from 0.7% in 2023.

Similarly, Verra,⁸ in addition to the combined VCS, launched new carbon capture schemes applicable to the AFOLU sector: Climate, Community & Biodiversity Standards, Forest Carbon Partnership Facility Program, and BioCarbon Fund Initiative for Sustainable Forest Landscapes Program. Boomitra⁹ has secured registration with Verra's Climate, Community, & Biodiversity Standards (covering SDG 1, 2, 3, 6, 8, 13, and 15). The targeted area under Boomitra's “Carbon Farming South of the Vindhyas” project in India is 88,000 acres (35,894 ha) which is expected to remove 400,000 Mg of CO₂e annually over a 20-year credit period. This project covers 23,000 smallholder farmers trained to practice climate smart land use approach. The second and third schemes mentioned above with Verra have yet to register any project which means that there is

⁶ <https://www.weforum.org/stories/2025/11/why-india-s-carbon-market-needs-a-price-stability-mechanism/>

⁷ <https://www.spglobal.com/commodity-insights/en/news-research/latest-news/energy-transition/110124-industry-sees-initial-carbon-credits-priced-around-10mtco2e-in-india>

⁸ <https://registry.verra.org>

⁹ <https://boomitra.com/projects/boomitra-carbon-farming-india/>

a vast potential for AFOLU in the carbon credit market under Verra. The latest release by Verra, VM0051 [119] enable rice production systems to be included in carbon credit projects with a detailed methodology for land use management (low input and low emission).

6 Key outcomes and lessons learned from carbon sequestration projects in India w.r.t carbon trading

India is a major player in the carbon market with a revenue of \$4010.5 M and market share of 0.8% in 2023, mainly from the compliance sector. With a projected growth rate of 43.2%, India is expected to have a revenue of \$49,448.6 M by 2030. The current biggest contributor is from the compliance sector but the voluntary sector will be the biggest sector in the coming years. To obtain a clearer picture on the carbon market across the world¹⁰ and where India stands, Table 2, shows the variability of revenue from carbon trading.

The Indian Carbon Market (ICM) is growing rapidly. The Indian government is preparing a new Compliance mechanism [120] under the ICM, in lines with CDM, with entity 'A' that can sell their carbon credit to Obligated Entity 'A', which can sell their carbon credit certificate (CCC) to Obligated Entity 'B' who can purchase the credits over the trading exchange. This scheme also provides banking of the CCC in the case of balance remaining after the compliance period, which can be utilized in the next compliance cycle or sold in the ICM. There is a provision for non-obligated entities that want to purchase the CCC voluntarily through the ICM. This scheme however, does not cover the AFOLU sector as of now, but this sector being energy dependent may be covered in the coming years as Obligated Entity 'A'. Agroforestry's role in carbon sequestration and its potential impact on India's environmental commitments is significant [121, 122]. Agroforestry systems in India have demonstrated the ability to sequester up to 20 Mg of carbon per hectare per year, highlighting their effectiveness in mitigating climate change [123]. Large-scale implementation of agroforestry can substantially contribute to India's NDCs under international agreements such as the Kyoto Protocol, REDD+, and the Paris Agreement, positioning agroforestry as a key strategy to achieve carbon neutrality and fulfill global climate goals [27, 57].

Table 2 Scenario of global carbon market and India's status in the global carbon market

Region	Asia Pacific	Middle East and Africa	Europe	North America	Latin America	India
Revenue (2024) (in M \$)	49,635.8	6509.1	573,657	11,280.4	3159.4	5444.7
Largest segment	Compliance	Compliance	Compliance	Compliance	Compliance	Compliance
Fastest growing segment	Voluntary	Voluntary	Voluntary	Voluntary	Voluntary	Voluntary
Projected Revenue (2030) (in M \$)	423,724.1	30,773.3	4,144,921.2	110,783.7	24,145.2	49,448.6
Growth Rate Projection (2025–2030) (in %)	43	29.6	39%	45.2	40.3	44.4

M= Million

¹⁰<https://www.grandviewresearch.com/horizon/outlook/carbon-credit-market>.

One of the critical outcomes of agroforestry projects in India is the increased income of farmers [57]. Farmers have benefited from selling timber, fruits, and bamboo, along with revenue generated from carbon credits [124]. The average income increments ranges from ₹40,000 to ₹75,000 per hectare per year from carbon credits, providing a significant economic incentive to adopt agroforestry practices [125]. This additional income enhances the financial stability of rural communities, and promotes sustainable land management.

Agroforestry significantly enhances environmental sustainability and promotes climate resilience [47]. These systems increase soil organic carbon by 15–25%, enhancing soil health and fertility [126]. Additionally, agroforestry reduces soil erosion and improves water retention, which are crucial for maintaining agricultural productivity in the face of changing climate patterns [47, 48]. The integration of trees enhances biodiversity, fostering a more balanced ecosystem that supports various flora and fauna. Furthermore, agroforestry systems have proven effective in helping farmers cope with climate change impacts, as increased shade cover and biodiversity improve farm productivity and resilience [127, 128].

With adequate policy support and market linkages, agroforestry-based carbon credit projects have the potential for significant scale-up [129]. Such expansion would provide farmers with billions of dollars in revenue, transforming rural economies and significantly contributing to carbon sequestration efforts [101]. These potential underscores the need for supportive policies and robust market mechanisms to facilitate the widespread adoption of agroforestry and to maximize its environmental and economic benefits.

7 Technological innovations in carbon monitoring and verification

Technological innovations have revolutionized carbon monitoring and verification in AFOLU (Fig. 2). Traditional carbon estimation methods are often costly, labor-intensive, time-consuming, and require extensive fieldwork, sampling, and laboratory analysis. Remote sensing and satellite technology offer a solution by providing real-time, large-scale monitoring of carbon stocks in forests, water bodies, croplands, and soils [27]. Direct access to tools for carbon accounting and datasets derived from remote sensing (RS) and geographic information systems (GIS) is critical for ensuring fair and accurate accounting of carbon credits, enabling transparent and verifiable carbon trading [120]. Advanced satellites such as NASA's Orbiting Carbon Observatory-2 (OCO-2) and the European Space Agency's Sentinel-2, remotely sense carbon stocks from agriculture, forestry, and other land uses. Hyperspectral imaging can detect minute details of the lithosphere and biosphere, enabling the detection of soil organic carbon in these areas [130]. Using high-end drones and cameras, Light Detection and Ranging (LiDAR) technology also captures imagery of forest canopies to estimate forest carbon, which is highly valuable in rugged terrain [131]. High resolution imageries can track wildfires, landscape degradation, droughts, and floods to detect patterns in carbon flux and leakage, while also allowing for the detection of forest cover loss and growing stock quality and quantity [132]. Platforms such as Google Earth Engine, Carbon Mapper, and Global Forest Watch aid this process. Processing these imageries to derive datasets and interpret carbon flux in agriculture, forestry, and other land uses extensively assists actors, policymakers, and

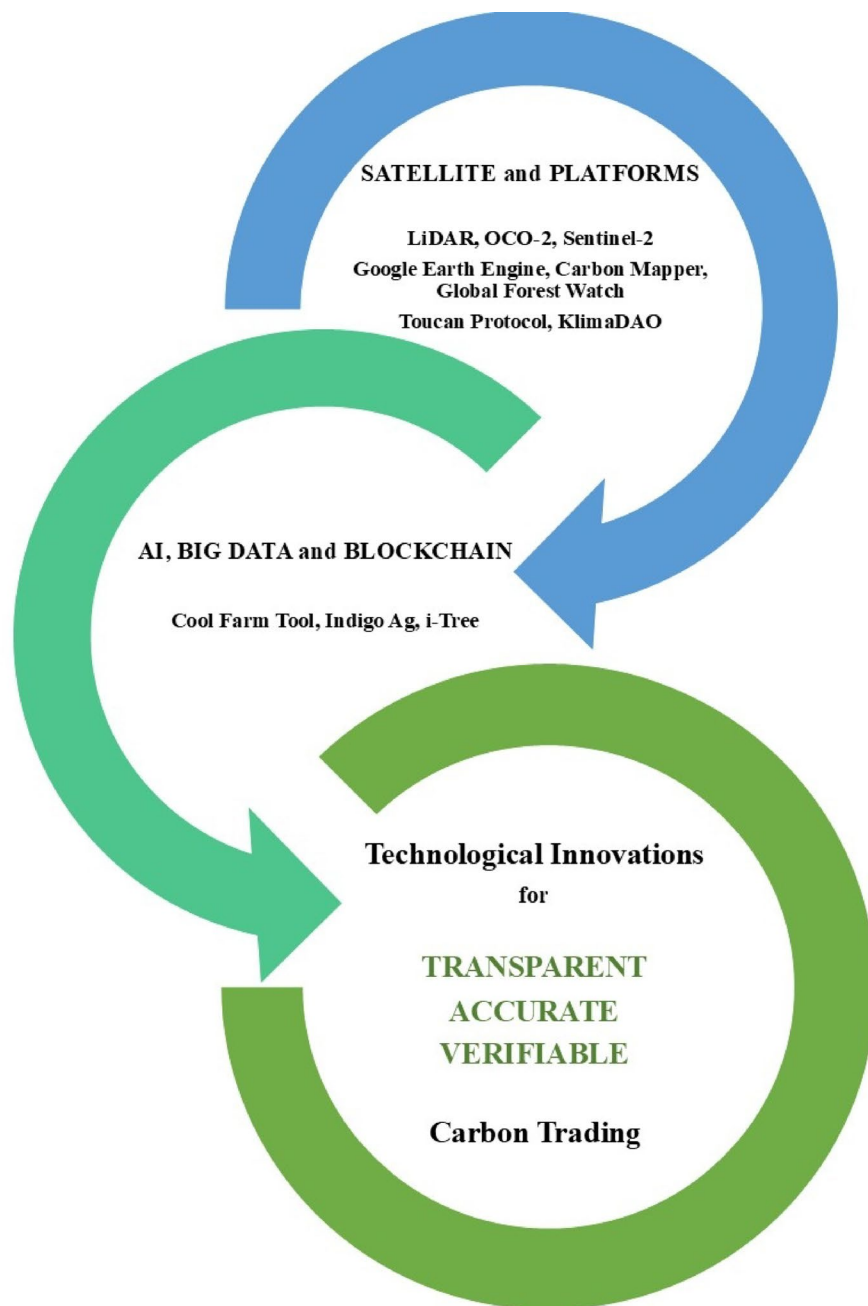


Fig. 2 Technological innovations in carbon monitoring and verification to enable a *just* carbon market

stakeholders in promptly responding to carbon leakages. These tools also effectively validate carbon offset projects and verify carbon sequestration claims [133].

AI and Big Data applications also play a crucial role in carbon accounting by relying on satellite imagery and existing carbon, weather, and climate datasets to precisely estimate carbon in agriculture, forestry, and other land uses [134]. AI applications such as Cool Farm Tool (CFT), Indigo Ag, and i-Tree are user-friendly tools that automatically estimate carbon emissions and sequestration, thereby determining net emission from land use. For instance, the CFT app requires information on water usage, synthetic fertilizers, pesticides, electricity, fuels used in transportation and machinery, tillage practices,

crop production and residue output, and farmyard manure to provide output in equivalent carbon units per land area. Moreover, the sequestration of carbon and other GHGs, water/precipitation interception, and groundwater recharge can be estimated with i-Tree [135]. These applications encourage agroforestry and cropland management practices, helping farmers track their emissions and estimate net emissions, which can be used as carbon credits, especially for VCMs by smallholder farmers. Big Data is used in agriculture carbon monitoring, verification, and fraud detection. Organizations such as the Regen Network and Pachama, along with Carbon Trust, employ deep learning to mine data from soil sensor data, farm management records, climate and weather data, imageries, and published works to verify carbon offsets and challenge fraudulent claims against agroforestry carbon sequestration projects [136]. Blockchain technology offers transparency in carbon trading, addressing issues such as fraudulent carbon credits, double accounting, and lack of transparency in carbon credit calculation and land use ownership [137]. Blockchain ensures that each credit is unique, verified, and certified by creating non-tamperable transaction histories for carbon credits. Platforms like Toucan Protocol and KlimaDAO have created tokens (digital assets) for unique carbon credits. Blockchain-based Smart Contracts operationalize the legal carbon trade only on verified carbon sequestration and ensure automatic payments to unique registered farmers based on verified soil carbon data [138]. Blockchain technology allows for a decentralized carbon marketplace as a digital token, facilitating direct rewards to sellers without third-party service providers, thus enhancing the full-price margin for farmers [139]. Real-time monitoring and auditing can be availed with Blockchain in platforms such as IBM Blockchain for Climate, which tracks the issuance and retirement of carbon credits from forestry-based carbon offsets [140].

The VM0051 [120] methodology also provides details for MRV and highly encourage the use of machine learning (ML) and artificial intelligence (AI) for accurate reporting, while also suggesting use of remote sensing (Google Earth) by VVBs for verification. It should be noted that, considerations for MRV with advance technologies may at present be a non-viable option for VCM participants as the prices per carbon credit is far lower than the cost involved, which ultimately leads to non-adoption by smallholders. Moreover, soil carbon measurement as well as complete biomass measurement is highly challenging owing to lack of harmonized standards. Another challenge in MRV lies in land governance, as many of the smallholders have unclear land tenure and are in small units which makes remote sensing very fragmented and complex [20, 141]. However, the Indian Council of Agricultural Research (ICAR) and the National Agricultural Research and Education System (NARES) are in the right direction to develop Tier-2 and Tier-3 emission factors for major agricultural practices to develop methodologies for MRV for major agricultural practices in India to upscale the potential of sustainable agricultural practices in participating in the VCM [106].

8 Effective carbon trading and sequestration in agroforestry and cropland: a policy roadmap

This section is prepared with the Indian scenario as a policy roadmap model (Fig. 3) by delving into the existing laws, policies, and schemes pertaining to agroforestry and cropland management, AFOLU and food security in India. Several policy approaches could be implemented to reduce carbon emissions and enhance carbon sequestration.

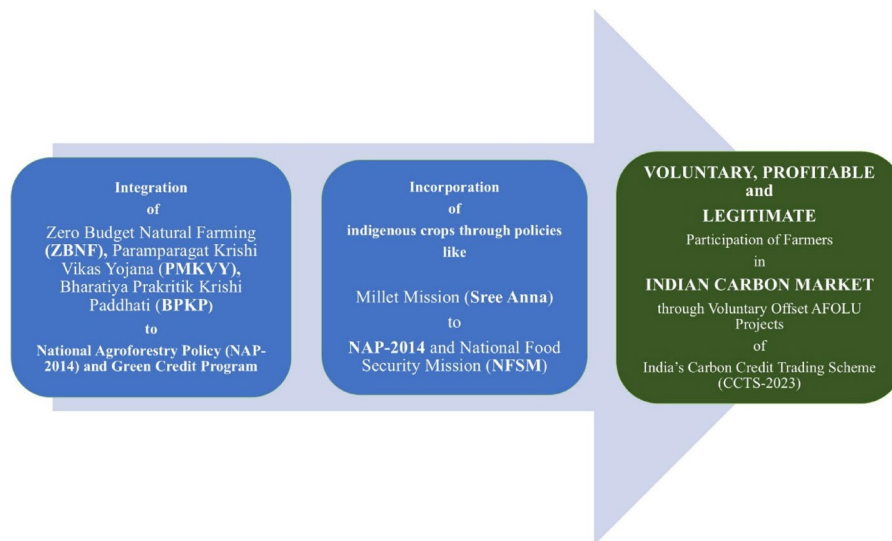


Fig. 3 Roadmap for a voluntary, profitable and legitimate participation of farmers in the Indian carbon market through agroforestry and cropland management relevant policy and schemes

Promoting indigenous agroforestry systems such as silvopasture, agri-silviculture, and home gardens can enhance carbon stock. These systems with inclusion of traditional practices in rural areas should be recognized and promoted by amending the National Agroforestry Policy (NAP), 2014 [142] and technical assistance to farmers (so as to optimize species selection, planting techniques, and management practices) thus ensuring that farmers gain carbon credits through traditional systems. Financial support, such as subsidies or grants, should be offered to farmers to establish and maintain traditional agroforestry systems, covering the initial investment costs and ongoing management expenses. India's agroforestry systems are highly diverse due to varied climatic conditions, with an estimated area of 13.75 million ha under agroforestry systems [125]. These land use can efficiently sequester carbon, and can also play a major role in diversification and sustainability, providing food security and reducing poverty [143, 144]. Bringing agroforestry and cropland under the Green Credit Program (2023)—which incentivizes environmental action through innovative market-based mechanism¹¹—can broaden the roles of these land-use systems in the VCM.

Traditional farming practices can be modified by conservation tillage and nutrient management to increase crop production and soil fertility [145]. Indigenous crops have low carbon footprints and require minimal chemical inputs, and policies like the Millet Mission (Shree Anna) and the National Food Security Mission (NFSM) should integrate carbon credit incentives and expand targets to promote their cultivation for carbon sequestration, while government programs including the Public Distribution System (PDS) and mid-day meal schemes prioritize their procurement to create stable markets, supported by increased research and development—using omics and gene editing tools—to enhance their yield and resilience, as exemplified by proso millet, a climate-smart, gluten-free, nutritionally rich grain with environmental benefits [146, 147].

Traditional methods such as zero-budget natural farming (ZBNF), organic composting, and crop residue management significantly enhance soil carbon content, and

¹¹ <https://www.pib.gov.in/PressReleaseframePage.aspx?PRID=1967476>.

schemes such as Paramparagat Krishi Vikas Yojana (PKVY) [148] and Bharatiya Prakritik Krishi Paddhati (BPKP) can be strengthened to increase farmers' eligibility for carbon credits. The budget and scope of PKVY and BPKP should be significantly increased to reach more farmers and provide more comprehensive support. The certification process for organic and natural farming practices should be streamlined, making it easier and more affordable for small farmers to obtain necessary certifications. Organic agriculture contributes to climate change mitigation through careful nutrient management and reduction in N₂O emissions [149]. Market linkages for organically and naturally grown produce should be facilitated, to ensure that farmers receive a fair price for their products. Certified organic products can cater to income options for farmers and promote climate-friendly practices worldwide [150]. Payments for ecosystem services (PES) can incentivize actions that increase the provision of services such as carbon sequestration [151]. Most importantly, the development of a national land use plan for optimization across carbon sinks, agriculture, and other conventional land use will ensure long-term outlook to balance land requirement and decarbonizing ambitions [152].

Small and marginal farmers practicing traditional farming should have easy access to carbon markets through simplified MRV. Local carbon aggregators and carbon finance institutions should be established to connect traditional farmers to the global carbon markets. Simplified and cost-effective MRV systems should be explicitly developed and tailored to traditional farming practices by leveraging remote sensing technologies and farmer-friendly data collection methods by the verification bodies. Australia's Carbon Farming Initiative provides lessons for regulatory design to reduce greenhouse gas emissions from agriculture [63]. A network of local carbon aggregators should be established to help small farmers bundle their carbon credits and access carbon markets collectively. Specialized carbon finance institutions should be created to provide loans and other financial products to support carbon sequestration projects in traditional farming systems. Quantifying carbon for agricultural soil management requires reliable, accurate, and cost-effective means of measuring carbon stock changes [153]. Innovative policies, such as a nitrogen credit system (NCS), could be implemented to incentivize the adoption of measures to reduce nitrogen losses from croplands [154].

Government policies should financially incentivize and integrate these practices into the Carbon Credit Trading Scheme (CCTS) 2023. Climate-smart agriculture technologies must incorporate the latest findings to mitigate potential risks and enable adaptation to stresses [155]. The carbon sequestration benefits of traditional land management practices should be explicitly recognized and valued within the Carbon Credit Trading Scheme. Direct payments or subsidies should be provided to farmers who adopt and maintain these practices based on the estimated amount of carbon sequestered. Community-based Forest management initiatives promoting sustainable harvesting practices and carbon sequestration in forest ecosystems should be implemented. It should be noted that incentives linked to short-term benefits have a higher adoption rate, but technical assistance is crucial for such incentives [93].

Community-led carbon projects based on traditional knowledge systems can enhance carbon sequestration. Indigenous peoples and rural communities practicing shifting cultivation, forest conservation, and indigenous water harvesting should be encouraged to generate carbon credits through these practices. Training and technical assistance should be provided to the indigenous peoples and rural communities to develop and

manage community-based carbon projects. Degraded land rehabilitation through agroforestry can address food, environmental, and livelihood security issues [156]. Communities should secure land tenure rights, incentivizing them to invest in long-term carbon sequestration projects. Transparent and equitable benefit-sharing mechanisms should be developed to ensure that communities receive a fair share of the revenue generated from carbon credits. Integrating agricultural landscapes with biodiversity conservation, secure land tenure, and benefit-sharing mechanisms is crucial for the success of community-based carbon projects [157].

The government should establish region-specific carbon measurement frameworks for traditional practices to ensure an accurate assessment of carbon sequestration and simplify the certification process, allowing more farmers to participate in carbon trading. Region-specific carbon measurement frameworks should be developed that consider each region's unique agro-climatic conditions and traditional farming practices. The “4 per mille” initiative aims to increase global soil organic matter stocks, but its feasibility varies by region [158]. Remote sensing technologies, artificial intelligence, and blockchain should be utilized to improve carbon monitoring, certification accuracy and efficiency. Precision farming enables agricultural management decisions to be spatially and temporally tailored, thereby reducing waste and environmental impacts [159]. The monitoring and certification processes should be transparent and accountable, with clear guidelines and procedures. Transparency and accountability are crucial for building trust in carbon markets and ensuring their effectiveness [160].

These policy approaches, grounded in the core principles of recognition, incentivization, accessibility, and community ownership, can pave the way for effective carbon trading and sequestration in India's agroforestry and cropland management while enhancing rural livelihoods and promoting environmental sustainability. The estimated area under agroforestry systems in India is 13.75 million ha, with the highest concentration being in the states of Uttar Pradesh (1.86 million ha), followed by Maharashtra (1.61 million ha), Rajasthan (1.55 million ha) and Andhra Pradesh (1.17 million ha) [121]. These practices are essential for the India to achieve Paris Agreement goals, as agriculture and land use contribute greatly to GHG emissions [62]. To mitigate nitrogen pollution from croplands in the future, innovative policies such as the NCS could be implemented to select, incentivize and, where necessary, subsidize the adoption of these measures [153]. Soil health and fertility are also major components of carbon sequestration, and soils worldwide are being increasingly affected by acidification, salinization, and pollution [154].

Various regions in India's landmass are covered under Boomitra's “Carbon Farming¹² in India” project (Verra ID:2889), which is mostly aligned with improved cropland management and targets the removal 400,000 Mg CO₂e annually. This project is seen as a significant leap in the VCM. Models can be developed such that policymakers and farmers (stakeholders) can easily understand the remuneration scheme for native tree species that can effectively sequester carbon and provide cost-effective means to participate in VCS scheme for smallholder farmers [161]. It is pertinent for nations and entities to carefully assess existing and up-to-date standards and practices to ensure quality under the Paris Agreement [162–164] and make the most of the carbon market dynamism.

¹²<https://boomitra.com/wp-content/uploads/2024/03/Carbon-Farming-in-India-CA-Assembly-Bill-No.-1305>.

9 Conclusion and recommendations

Carbon trading and sequestration in agroforestry and cropland management are crucial strategies for mitigating climate change, particularly in a country like India, which has a vast agricultural landscape and a rich tradition of farming practices. To ensure the effectiveness of these strategies, it is essential to integrate traditional farming practices and native crop varieties into carbon-efficient models. India's wealth of land use practices and varieties holds immense potential for carbon sequestration and emission reduction, which can be unlocked through strategic integration into national policies and carbon credit mechanisms, leading to significant environmental and economic benefits while empowering rural communities. Several core principles must be adopted to exploit this potential. First, the inherent carbon sequestration benefits of traditional farming practices and native crop varieties must be acknowledged and validated using rigorous scientific methods and standardized measurement methodologies. Second, financial and technical support must be provided to farmers who adopt these practices to make them economically viable and attractive. Third, small and marginal farmers, often custodians of traditional knowledge, must have easy access to carbon markets and related benefits. Finally, community-led carbon projects that build upon traditional knowledge systems should be fostered to empower local communities to manage and benefit from their natural resources.

This comprehensive strategy requires a collaborative effort involving government agencies, research institutions, NGOs, and, most importantly, farmers. Through strategic policy integration, targeted investments, and community empowerment, India can harness its power of tradition to build a sustainable and resilient future. Finally, India's participation in the Tropical Forest Forever Facility¹³ and the Green Credit Programme¹⁴ may bring a paradigm shift to amend and improvise existing policies to complement forest finance programs, including REDD+ and voluntary carbon markets. Thus, a combination of decisive policy at the national, catalysed by civil society organizations, can shift the trajectory from looming climate tipping points to positive tipping point opportunities—a shift that can only be achieved through the effective participation of small-holder farmers and carbon aggregators. The following five recommendations are made to achieve a farmer friendly carbon market.

- i. A multi-pronged approach is crucial for unlocking the full potential of agroforestry and cropland management for carbon sequestration. Policymakers should integrate these practices into national climate change policies, offering financial incentives such as subsidized loans and grants for tree plantation, irrigation, soil restoration, and carbon credit registration. Streamlining the carbon credit registration process and establishing direct buyer-farmer linkages will strengthen market access.
- ii. Training and extension services are essential to educating farmers on agroforestry, carbon sequestration techniques, and carbon credit trading. Governments should foster public–private partnerships and integrate carbon sequestration into National Climate Action Plans.
- iii. Relevant authorities should accelerate the digitization of land records, with due safeguards as outlined in the National Forest Policy (1988) and the Forest Rights Act

¹³<https://tfff.earth>.

¹⁴<https://www.moefcc-gcp.in>.

(2006) of India. This will facilitate identification of suitable lands for carbon projects. Similar exercise can be done in countries with similar agricultural and land use management scenario.

- iv. Leveraging technology, such as satellite imagery and blockchain, can simplify MRV. Research on high-carbon sequestration crops and region-specific agroforestry models is vital.
- v. Farmers should adopt agroforestry, register carbon credits, and engage in carbon markets. Investors and corporations can invest in carbon offset projects and promote sustainable agricultural supply chains. These efforts will reduce carbon emissions, enhance food security, promote sustainable land management, and contribute to global climate action goals.

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TS—Conceptualization, reviewing, writing main draft and editing; LP—Editing.

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Competing interests

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